

MODULAR SUBSURFACE IMAGING TOOL

Discovering Hidden Structures Through Accessible Electrical Imaging

Technology Student Association
Engineering The Tools For Scientific Discovery

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PROBLEM ANALYSIS

Introduction To The Engineering Challenge

Scientific discovery is the foundation of human advancement. Its influence is spread through every part of our world and every material, tool, or object manufactured. The Merriam-Webster Dictionary defines discovery as “the act or process [of]... obtaining sight or knowledge for the first time.” Therefore, scientific discovery is the process of obtaining knowledge using scientific means. Because of the importance of this process, one of the United States National Academy of Engineering’s (NAE) Grand Challenges for Engineering is to Engineer Tools for Scientific Discovery.

However, despite the simplicity of the challenge statement, the job of engineering those tools is complex. The goal of engineering these tools is to enable scientific discovery- a goal that is broad in scope. The NAE has defined multiple important areas that engineers can focus on to fulfill the challenge. Although the possible ways to fulfill this challenge are nearly infinite, two broad areas stand out.

The first of these lies in biology. As the NAE states, “Biologists are always seeking, for instance, better tools for imaging the body and the brain... Scientists still have much to learn about the relationship of genes and disease, as well as the possible role of large sections of our DNA that seem to be junk with no function, left over from evolution.” Biology provides an amazing example of where scientific discovery occurs, as much of the human body is still unknown, with more being found out about the processes behind our function every day. Several other of the grand challenges for engineering are directly correlated to biology, including reverse engineering the human brain and engineering better medicines, both of which are impossible without the help of machines engineered with the intent of being tools for scientific discovery.

The second is underground discovery, with fields like paleontology and archaeology. Over 100 years ago, paleontologists used shovels, chisels, and basic environmental context to discover fossils. They would travel to environments that were ideal for creating fossils and carefully excavate. Surprisingly, modern methods are mostly the same; finding fossils still mostly depends on luck and environmental databases. This static nature has plagued paleontology for centuries. Only recently have researchers begun experimenting with new technology. Technologies such as ground penetrating radar (GPR), LiDAR, electrical resistivity tomography (ERT), and more recently, quantum sensor gravimeters. While these tools are advanced, they are expensive and still far from ideal. This is the problem faced: how can modern technology be advanced beyond imagination, yet paleontologists are stuck using primitive techniques or working with temperamentally experimental technology? This is unheard of for fields supposedly making cutting-edge discoveries. As a result, fossil discovery has fallen far from the modern cutting edge. Solving this epidemic will be challenging, as experiments using technology like GPR in paleontology settings are few and far between, another common problem.

PROBLEM ANALYSIS

History of Scientific Discovery

Scientific discovery has always been a part of the history of humankind. It is the vehicle that has transported humans from their first inception to the present. From the first rock used as a weapon, to fruit being discovered as edible, to the first utilizations of fire. Every step of the path of human history has been a result of scientific discovery. A worn path that will not change. The way forward for society is the same as it has been since the first civilization. Creation, ingenuity, and daring. All components are required to step into the unknown; a requirement for discovery.

Enhancing the system of scientific discovery is in the best interest of the entire population. For our society to evolve, we must keep moving onward. Stepping into new and foreign territory using science. The way we allow this to happen at the highest level is to have the right equipment to do so. For discovery to be made, new tools need to be created, improved, and streamlined. To establish new ideas at the frontier of nature.

As the National Academy of Engineering's Grand Challenges of Engineering establishes, "Engineering's success in finding answers to nature's mysteries will not only advance the understanding of life and the cosmos, but also provide engineers with fantastic new prospects to apply in enterprises that enhance the joy of living and the vitality of human civilization." In essence, there is an eternal problem. The tools available for the sciences will never be perfect, an enhancement or adjustment can always be made.

Sometimes, the solution to this age-old struggle lies in providing the right person with the right equipment. Not at the level of advancement for said equipment. Creating tools that are more accessible and usable for the common society often allows for discoveries to be made that otherwise would be left unturned.

If strides are not consistently made towards developing more advanced tools for scientific discovery, society's advancement will become stagnant. Without adequate equipment, scientists can't conduct necessary cutting-edge research, visionaries can't make profound discoveries, and all that is still unknown will remain as such. Society will be bottlenecked by a surplus of inadequate technology. Thus, the persistent challenge for engineers to create the tools for scientific discovery is crucial for the health and development of society.

PROBLEM ANALYSIS

Problem Background

Subsurface imaging is the process of surveying the ground underneath, using various methods to have a better understanding of the composition of the ground and the density of the materials.

This information is important because these simple details can tell a lot about what is going on underground without using the means of excavation. Because excavation consumes a lot of time and resources. It can also be used for a multitude of applications such as infrastructure, disaster response, and archaeology. Which can all lead to scientific discovery in its own right.

Conversely, there are currently multiple problems with the current systems that are being used. One problem, for example, is the cost of the systems themselves. Ground Penetrating Radar [GPR] systems reaching up to tens of thousands of dollars. In addition to this, they can also take up a lot of space, like Electrical Resistivity Tomography [ERT], for example, which can span over 50 yards and is also quite expensive.

PROBLEM ANALYSIS

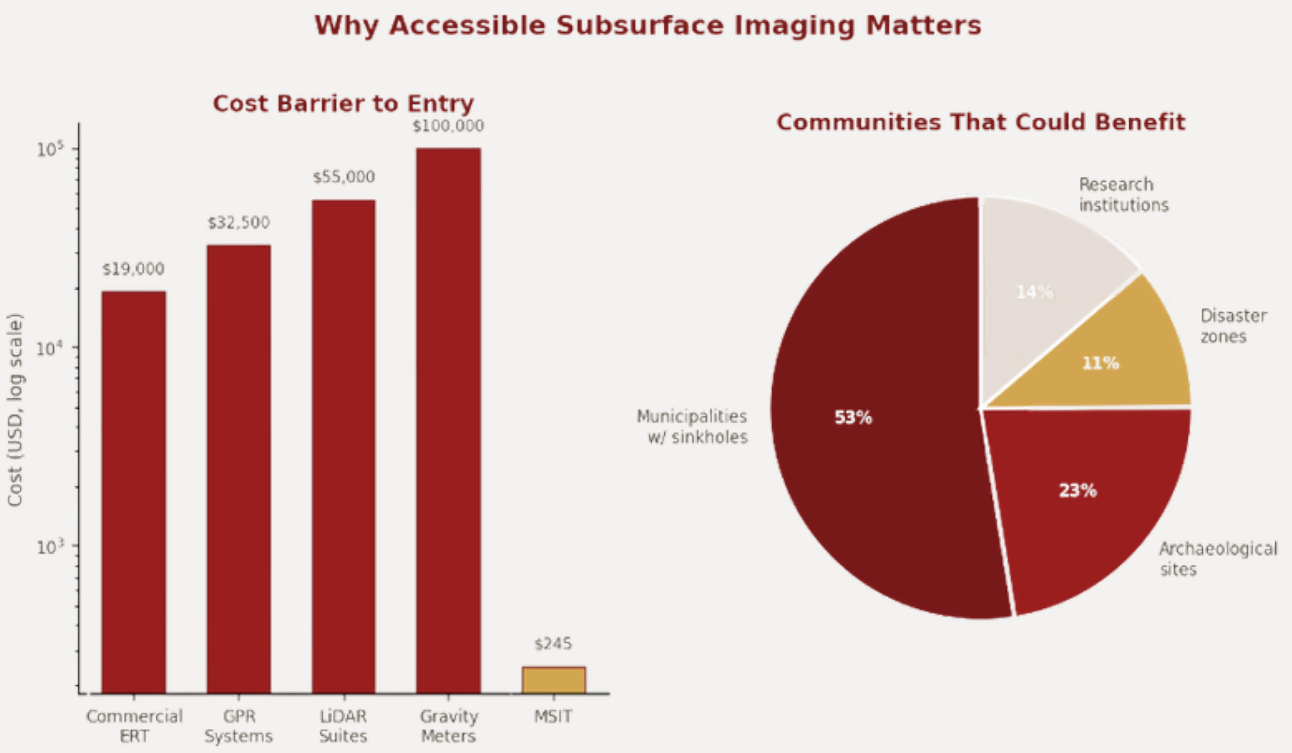
Existing Technology Analysis

Today, an option for an affordable and accurate subsurface imaging tool is not readily available. Finding the balance between capabilities, accuracy, price, and user friendliness is going to be the main problem that will have to be solved for the MSIT to accomplish its goal.

Technology	Cost (USD)	Accuracy	Accessibility	Depth Range
ERT (Commercial)	$8,000–$30,000	High	Moderate	1–20 m
GPR Systems	$15,000–$50,000	Very High	Low	0.5–5 m
LiDAR Suites	$50,000+	Moderate	Very Low	Surface only
Quantum Gravimeters	$100,000+	Low	Very Low	Unlimited
MSIT (This Project)	**< $245 ✓**	High	Very High	1–10 m

SOLUTION ANALYSIS

Why This Problem Matters



Technology	Cost	Relative Cost (visual)	Multiplier	Status
Commercial ERT (avg)	$19,000		**80x**	Commercial only
GPR Systems (avg)	$32,500		**160x**	Commercial only
LiDAR Suites (avg)	$55,000		**220x**	Commercial only
Gravity Meters	$100,000		**400x**	Commercial only
MSIT (This Project)	**$245 ✓**		**1x (baseline)**	**97% cost reduction**

SOLUTION ANALYSIS

Impact Analysis

WITHOUT MSIT	WITH MSIT
✗ Commercial ERT: $8,000–$30,000	✓ Total build cost: $150
✗ GPR System: $15,000–$50,000	✓ 100% commercially available parts
✗ LiDAR Suite: $50,000+	✓ USB-powered & battery portable
✗ Requires trained specialists	✓ Deployable in under 2 hours
✗ Multi-day deployment setup	✓ Open-source firmware (GitHub)
✗ Limited to well-funded orgs only	✓ Accessible to NGOs & schools
Cost: $8,000 – $100,000+	→ **Cost: $245 (97% reduction ✓)**

SOLUTION ANALYSIS

Potential Applications

The MSIT is an extremely versatile tool, with it having the ability to read a signal on any material due to all materials having electrical resistance. This means that it can be used for a plethora of people's needs. With user friendliness in mind, the MSIT is able to not only allow these people to make their own discoveries, but it readily facilitates it in a way that is more comfortable for the user.

Application Domain	How MSIT Helps	Key Use Cases
Archaeology	Locate buried artifacts and historical structures without excavation. MSIT maps subsurface resistivity anomalies indicating organic material, voids, or stone foundations.	*Buried artifacts, foundations, historical sites*
Agriculture	Map soil moisture distribution and soil layer variation across fields for precision irrigation and crop planning.	*Moisture mapping, drainage assessment, soil profiling*
Disaster Response	Rapidly assess post-earthquake or post-flood ground safety by detecting subsurface voids, liquefaction zones, or unstable fill.	*Sinkhole detection, post-earthquake safety, flood damage*
Infrastructure	Detect subsurface voids, cavities, and soil anomalies beneath roads, bridges, and building foundations before failure.	*Road voids, utility location, foundation assessment*
Education	Provide high school and university geophysics labs with an affordable, hands-on ERT instrument at 1/100th the cost of commercial systems.	*University labs, STEM outreach, field courses*

INFORMATION GATHERING

Research Questions & Findings

Research Question	Finding	Design Impact
Can radar penetrate clay soils?	No — signal absorbed by moisture and clay minerals.	Abandoned FMCW radar; chose ERT instead.
What is the optimal current source circuit for ERT?	Howland Current Pump provides stable, ground-referenced constant current output.	Implemented LM324-based modified Howland pump.
Which ADC resolution is needed to resolve soil resistivity signals?	≥16-bit needed to resolve µV-level differential voltages across electrodes.	Selected ADS1115 (16-bit, I²C interface) × 2.
Can instrumentation amplifiers reject 60 Hz interference?	INA333 achieves >100 dB CMRR; eliminates mains noise.	INA333 chosen over INA128 for CMOS inputs and zero-drift.
What electrode spacing gives best depth vs. resolution tradeoff?	0.2m spacing → ~0.4m investigation depth (Wenner formula: depth ≈ a).	8 electrodes at 0.2m spacing on parallel array
Can the $250 cost target be met with commercial off-the-shelf parts?	Yes — full BOM = $245.14 including contingency.	All components sourced from Digi-Key / Amazon.

INFORMATION GATHERING

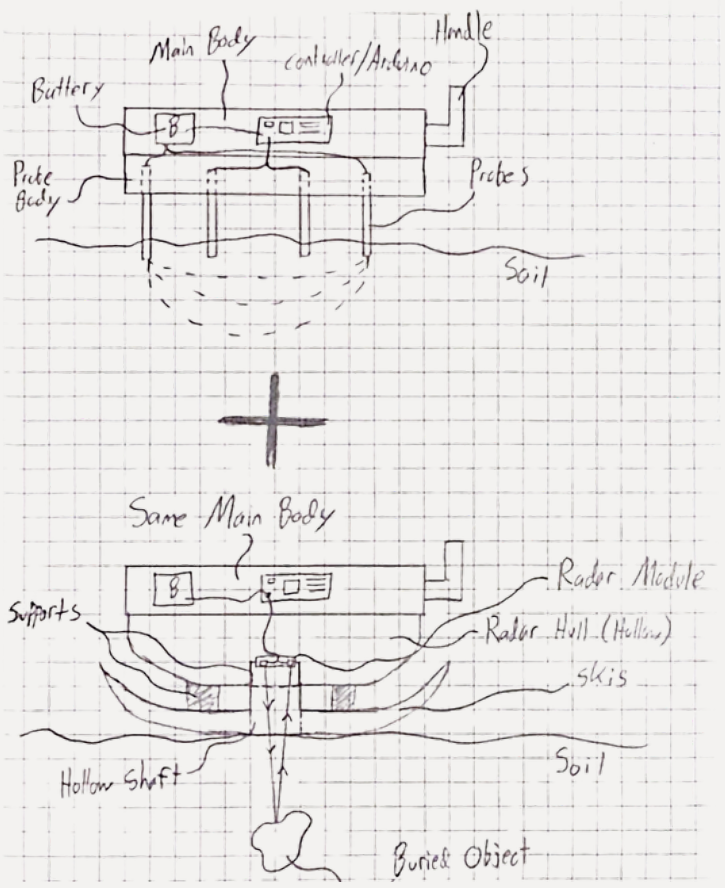
Component Selection

Component	Alternatives Considered	Final Choice	Selection Rationale
Microcontroller	*Arduino Uno, Raspberry Pi Pico, STM32*	**Teensy 4.0**	600 MHz ARM Cortex-M7; native USB HID; 3× I²C masters; 256KB RAM — no bottlenecks for 28-measurement scan cycle.
Instrumentation Amplifier	*INA128, AD620, LT1167, INA125*	**INA333**	CMOS input bias current (picoamps) vs BJT (nanoamps); zero-drift/chopper architecture eliminates 1/f noise; single-supply.
Analog-Digital Converter	*MCP3208 (12-bit), ADS1015 (12-bit), internal Teensy ADC*	**ADS1115 × 2**	16-bit resolution; 860 SPS; programmable gain amplifier (PGA); I²C address configurable (0x48/0x49); <$4 each.
GPIO Expander	*MCP23S17 (SPI), PCF8574 (8-bit I²C)*	**MCP23017**	16-channel I²C (not SPI — saves Teensy SPI pins); address 0x27 via onboard pull-ups; controls both CD74HC4067 muxes.
Multiplexer	*CD4051 (8-ch), ADG1206, 74HC4051*	**CD74HC4067 × 2**	16-channel per IC; 5V logic compatible; low Ron ≈ 70Ω; 4-bit select (S0–S3) driven by MCP23017.
Boost Converter	*XL6009, XL7015, MT3608*	**MT3608**	Compact 1.2 MHz PWM; 2–28V input → up to 28V; caps at ~10.5V (acceptable for demo); <$1 unit cost.
Current Source Circuit	*Standard Howland, Improved Howland, VCCS variants*	**Modified Howland (LM324)**	Grounded load configuration; 4× 10kΩ resistors hand-matched to ±0.1% tolerance; stable ±1% output current.
Electrodes	*Copper pipe, steel rods, stainless rod, graphite*	**Brass Rod (6mm × 30cm)**	Corrosion-resistant; low contact resistance; food-grade alloy; cost-effective; mechanically robust in soil.

Gold = final selected component. All alternatives were evaluated against cost, availability, electrical performance, and compatibility with the I²C bus architecture.

POSSIBLE SOLUTIONS

Frequency-Modulated Continuous Wave Radar + ERT



The final solution to designing the tools for scientific discovery devised was a geophysical sensor with a swappable base that utilize's Frequency Modulated Continuous Wave radar. The central body is the brain, housing the Arduino, battery pack, and WiFi/Bluetooth enhancement. A dovetail slide allows the two unique bases to be swapped. One base, the ERT rail, is a female dovetail block with holes large enough to have four probes inserted. The probes would then be plugged into the Arduino via alligator clamps, and an injection current would be sent through the two outside probes. While the inner probes measure the voltage difference. The second swappable base is a hollow sled-based device that is attached below another female dovetail block. The sled is home to an empty center column for the radar module to accurately derive readings, along with a long base to disperse force. Separate wiring would connect the module to the "brain".

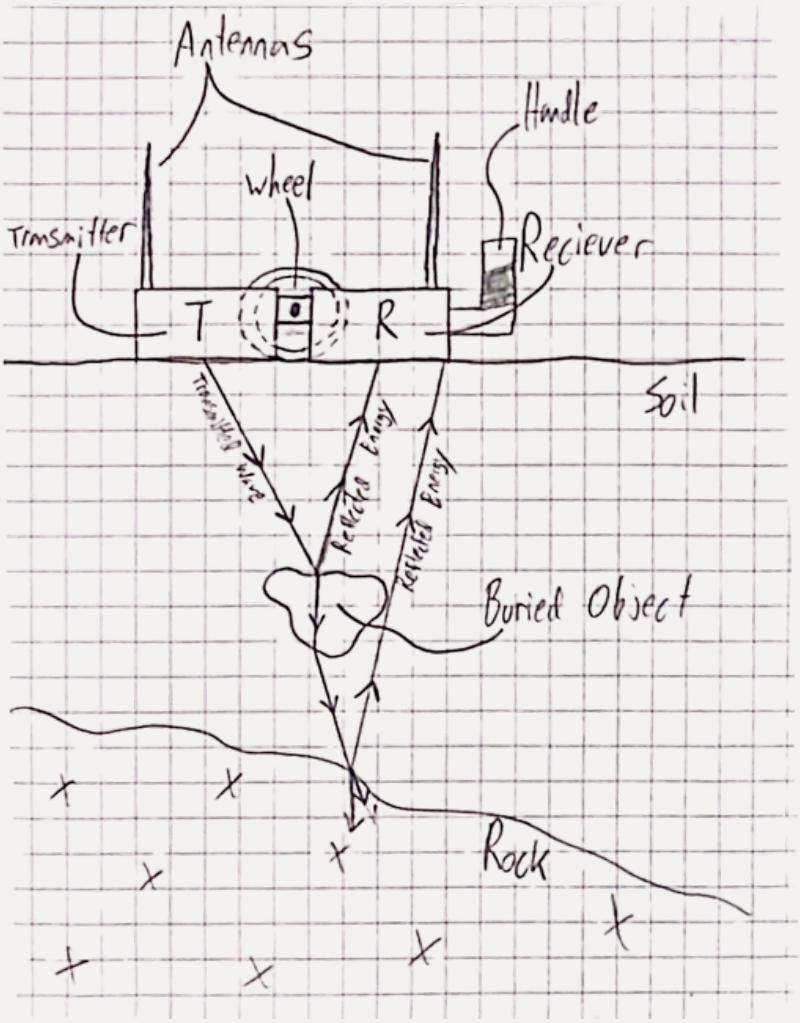
Ideally, the device would be powered by a battery pack, allowing for movement ease and portability. Both bases will be powered by pressing a button on a paleontology device, where data will be automatically uploaded. But the Arduino R3 has neither Wifi nor Bluetooth capabilities, therefore an additional module is required.

This idea stemmed from the research conducted on the techniques used in paleontology and archaeology. We found that most techniques used were clunky, hard to set up, and inefficient. Thus, our idea developed into combining two discovery techniques into an easily portable and exchangeable handheld device.

POSSIBLE SOLUTIONS

Ground Penetrating Radar

The second solution considered was a wheeled, ground-penetrating radar device. This device utilizes a transmitter, receiver, and antenna to send an electromagnetic pulse into the ground and read the results sent back off surfaces. The results would easily indicate irregularities and general shapes. The antennas are responsible for both emitting and receiving radar waves. It scans the subsurface, detecting materials and structures by bouncing radar signals back to the control unit, the brain that manages the waves received. The wheel allows for smooth travel among many different terrains.



The idea came from searching for technologies that would allow for a clearer mapping of underground terrain compared to the electrical resistivity tomography design in Solution 1. Once we established the plausibility and safety of sending radar waves into the floor, we moved onto the research of current designs in use today.

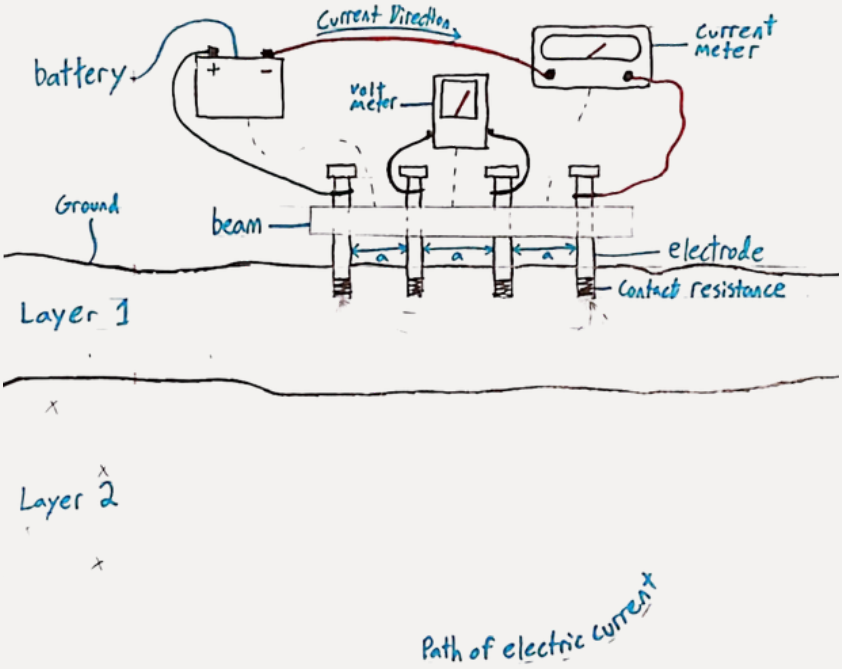
Ideally, paleontologists/ archaeologists would roll this device along zones they were investigating and receive data that allows them to cleanly excavate areas with artifacts without damaging said artifacts. The received signal would be sent to a Bluetooth-connected display screen, automatically generating graphs that clearly indicate any anomalies.

A clear flaw with this design is its ability to traverse through rockier terrains. Though the wheel(s) would function correctly when faced with sand or gravel, when it comes to rocky/hilly terrain the model would be unable to efficiently mobilize. But, as the sensors would require close proximity to the ground, there was no clean way to have the sensors far enough away from the ground while also having the mechanical capability to traverse rocky/bumpy terrain.

POSSIBLE SOLUTIONS

Electrical Resistivity Tomography

Not only are there many fields where scientific discovery tools are necessary, but within each field there are often hundreds of different tools used. So it was crucial the field was picked first, and archaeology/paleontology were chosen. The first solution to this problem was an electrical resistivity tomography board. The design is a board that holds four probes stagnant while being connected to a battery, current-meter, volt-meter. As electricity is sent through the two outside probes, the current-meter allows us to see how much power the battery is sending out. Then, the two inside probes receive the voltage from the outside probes after it goes through the ground. The volt-meter gives us a reading of the voltage after the electricity is sent through the ground.



The idea was that the electricity reading received and documented would show what the material makeup of the area under surveillance was. Once the data was received, the voltage values could be compared to a database to establish exactly what kind of material was beneath them. Giving paleontologists a stronger idea of what the area they were digging looked like, improving efficiency and reducing excavation damage. And, if there were a fossil or underground structure, the voltage readings would demonstrate that there was an anomaly.

This solution is far from perfect though. Different arrays of the four probes enable different levels of data to be collected. Like the Wenner Array depicted, inner 2 electrodes measure potential difference (V), outer 2 electrodes are used to pass current through. But we realized that the ability to conduct research using different kinds of array's would be invaluable. Thus, while the design would function, it lacks mechanical and programming capability depth.

POSSIBLE SOLUTIONS

Decision Matrix

Criterion	Weight	FMCW Radar	GPR	ERT (MSIT)
Cost ≤$150	25%	3	1	5
Clay/Wet Soil Penetration	20%	1	2	5
Quantitative Resistivity Output	20%	1	1	5
DIY Complexity (higher=simpler)	10%	3	2	3
Effective Depth Range	10%	2	3	4
Data Quality / Accuracy	10%	2	4	4
Portability	5%	4	3	5
WEIGHTED TOTAL (max 5.00)	**100%**	**2.05**	**1.90**	**4.60**

Score key: 5=Excellent 4=Good 3=Adequate 2=Poor 1=Fail

SELECTED SOLUTION

MSIT ERT Overview

After evaluating our three main concepts with variables that we find important for the goal of giving people affordable access to subsurface imaging tomography, our group selected the Electrical Resistivity Tomography solution. We found that this solution was the most effective when prioritizing price, quality of data, ease of use, portability, and versatility. This allows us to best solve the challenge of Engineering Tools of Scientific Discovery by picking the best solution that enables anybody to be able to analyse soil conditions and irregularities without paying upwards of 30,000 dollars.

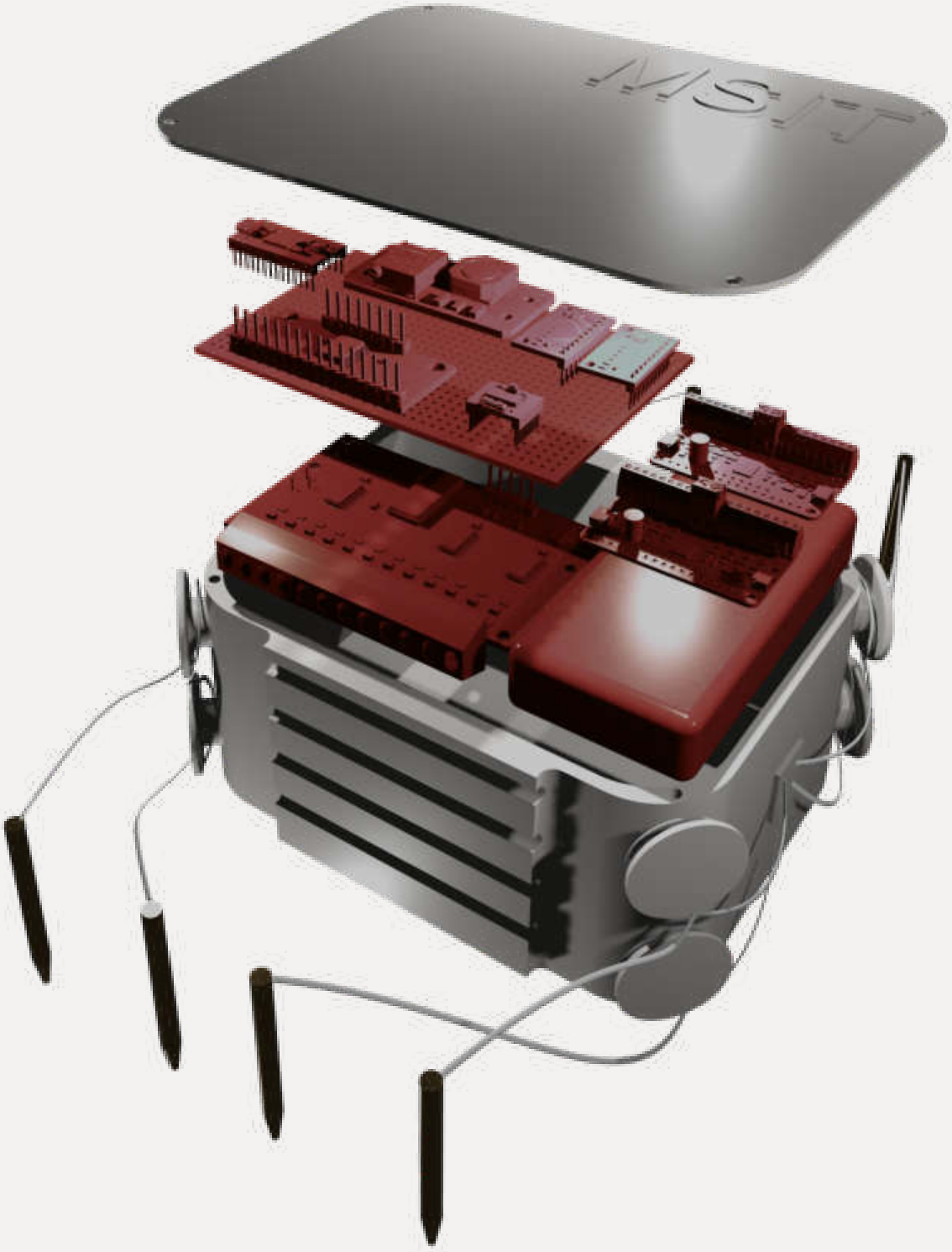
That solution is the gradient array. The premise of this solution is that the outer probes are grounded and power and they retain the same job throughout the readings without changing. Unlike the Wenner array. Whilst the inner probes are reading the voltage drop, the resistance of the soil. This can tell you the difference in the soil across the area that is being read. Because of this, the person who is collecting data should be easily able to detect any anomalies in the soil.

By using previous information about the resistivity of soil types, you can calibrate the probes to be able to correctly read the differences in the soil. By doing so, you can detect ideal areas for fossil formation, air pockets, condensed material, etc. The benefit of this solution is also that it does not span over fifty yards, much like ERT. It is also less complex and easier to troubleshoot and calibrate if something were to go awry.

The other benefit of the simplicity is that it can be combined with other subsurface imaging tools and can thus be modular. Then, more data can be collected, and the accuracy and precision of the system increase. Allowing for accurate, accessible data.

SELECTED SOLUTION

System Architecture



SELECTED SOLUTION

Electrode System

Material Specifications

Parameter	Value / Notes
Electrode material	Brass 360 (60/40 Cu/Zn alloy)
Diameter	6mm round rod
Length per electrode	300mm (30cm)
Quantity	8 electrodes per survey array
Electrode spacing (a)	500mm (0.5m) — uniform Wenner geometry
Total array span	3.5m (7 gaps × 0.5m)
Carrier rail	PVC conduit, 3.5m length, 8× slots at 0.5m
Connection method	Alligator clip → 3.5mm banana jack on enclosure
Contact resistance (wet)	< 5Ω typical in moist soil
Contact resistance (dry)	50–500Ω (apply saline water to improve contact)
Effective depth range	~0.5m – 3.0m (function of electrode spacing and array)

Electrode Roles in Array

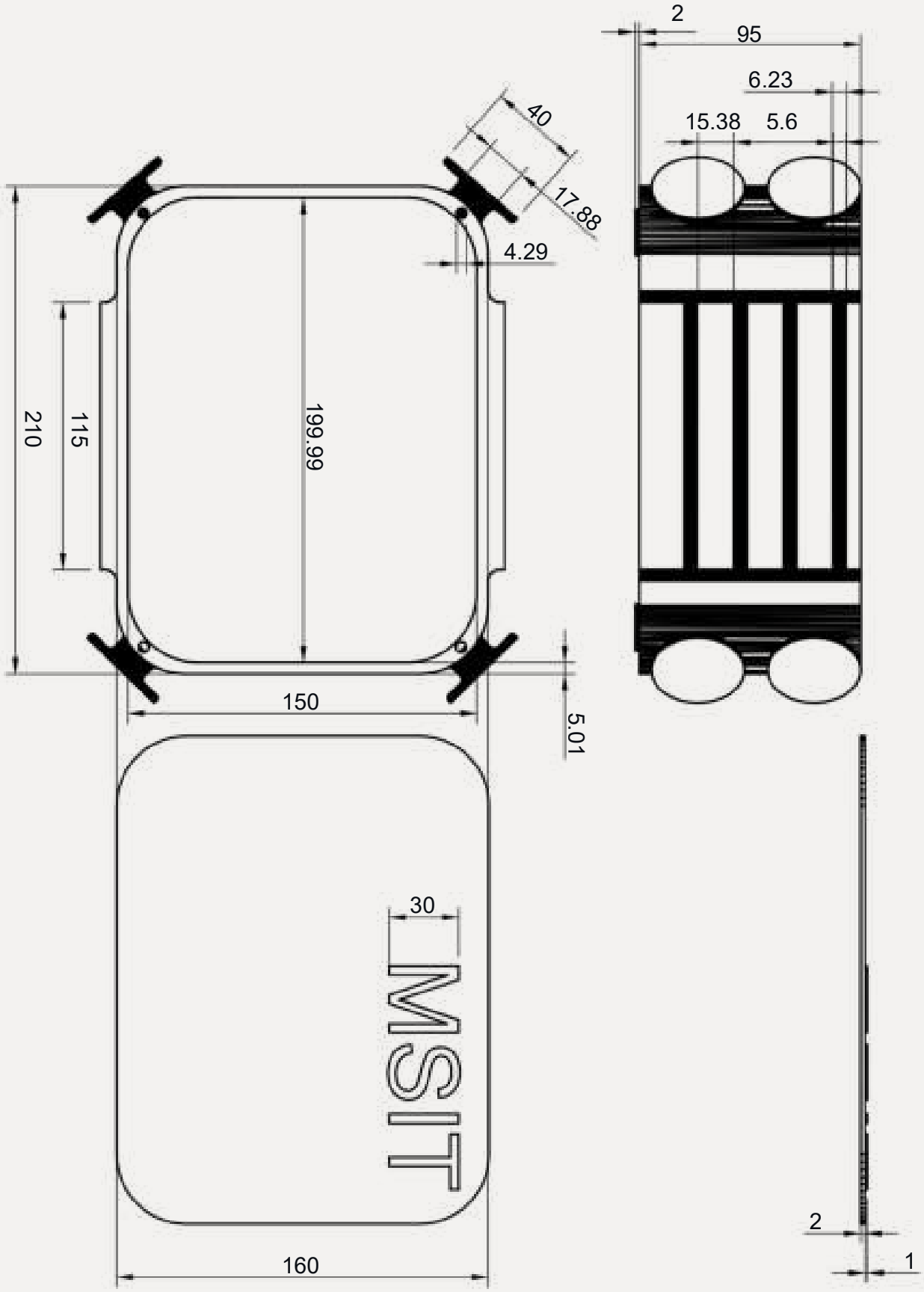
Electrode	Role
E1 (C1)	Current injection — positive terminal
E2 (P1)	Voltage measurement — positive sense
E3 (P2)	Voltage measurement pair (in basic 4-el. mode)
E4–E7	Additional sense pairs for gradient array mode
E8 (C2)	Current injection — return terminal

Array Physics & Resistivity Formula

Parameter	Value
Wenner-α spacing	All pairs at uniform distance a = 0.5m
Investigation depth	≈ a for Wenner (0.5m min); deeper with wider pairs
Geometric factor K	$K = 2\pi a = 2\pi \times 0.5 = 3.14159$ m
Apparent resistivity	$\rho a = K \times \Delta V / I = 2\pi a \times \Delta V / I$
Number of measurements	28 unique 4-electrode combinations from 8 electrodes
Measurement cycle time	~2 minutes for full 28-point profile

SELECTED SOLUTION

Enclosure Design



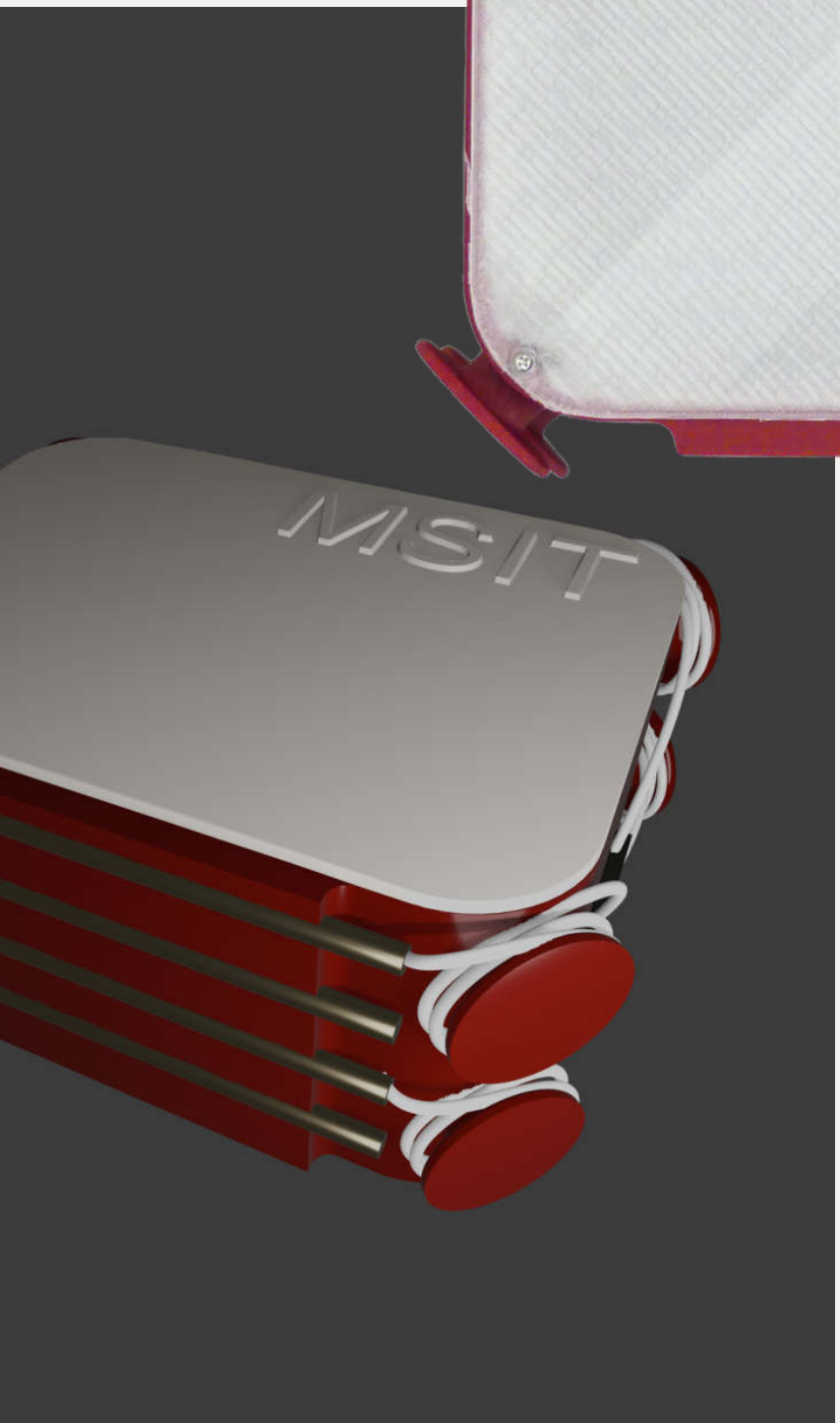
SELECTED SOLUTION

Enclosure Design



3D Printed Model

Rendered on Blender



SELECTED SOLUTION

Electrical Schematic

POWER SUPPLY			
Parameter	**Value**	**Parameter**	**Value**
Component	9V Battery + MT3608 Boost	**Output**	10–12V regulated rail
Input range	2–28V (MT3608 spec)	**Capacitors**	100µF + 10µF decoupling
Efficiency	~90% at <200mW load	**Notes**	Caps at ~10.5V — acceptable for demo
HOWLAND CURRENT PUMP (LM324)			
Parameter	**Value**	**Parameter**	**Value**
Op-amp	LM324 quad (uses 1 of 4)	**Topology**	Modified Howland
R1=R2=R3=R4	10kΩ, matched ±0.1%	**Output current**	~1mA (V+/R4)
LM324 supply	10V from MT3608	**VREF input**	2.5V (from LM324 buffer)
SWITCHING — MCP23017 + CD74HC4067			
Parameter	**Value**	**Parameter**	**Value**
MCP23017 addr	0x27 (A0=A1=A2=1)	**I²C speed**	400 kHz (fast mode)
Mux A (CD74HC4067)	Current injection select	**Mux B (CD74HC4067)**	Voltage sense select
Ron (on-resistance)	~70Ω per channel	**Logic level**	5V (compatible with Teensy 3.3V + level shift)
SIGNAL CHAIN — INA333 → ADS1115			
Parameter	**Value**	**Parameter**	**Value**
INA333 gain	G = 1 + 99,900/Rg = 1001	**CMRR**	>100 dB
REF pin	2.5V via LM324 buffer	**Input range**	±ΔV/G ≈ ±few mV → ±few V out
ADS1115 PGA	±256mV (code 5)	**Sample rate**	128 SPS (4 averaged)
MICROCONTROLLER — Teensy 4.0			
Parameter	**Value**	**Parameter**	**Value**
MCU	ARM Cortex-M7 @ 600MHz	**RAM / Flash**	256KB / 2MB
I²C pins	SDA=18, SCL=19	**USB**	Native USB HID / Serial
Scan cycle	28 electrode combos	**Output**	USB Serial → CSV

All five subsystems communicate via the I²C bus. The Howland pump runs continuously; the MCU selects electrode pairs via MCP23017→mux, waits 10ms for settling, then reads both ADS1115 channels.

SOLUTION ABSTRACT

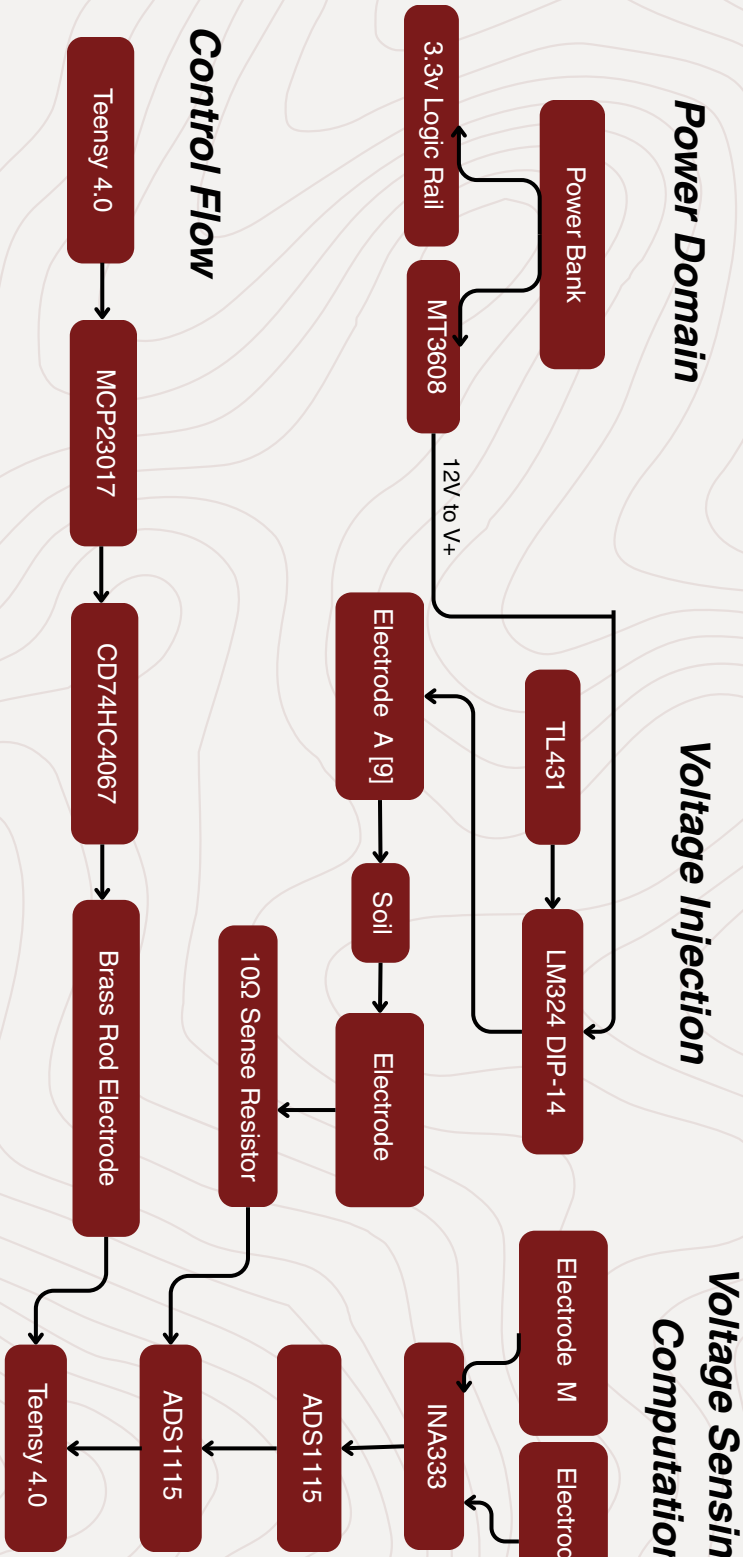
Cost Analysis

Component	Qty	Unit Cost	Total	Category
Teensy 4.0 Microcontroller	1	$29.00	**$29.00**	*MCU / ICs*
INA333 Instrumentation Amplifier	1	$8.50	**$8.50**	*MCU / ICs*
ADS1115 16-bit ADC	2	$3.50	**$7.00**	*MCU / ICs*
MCP23017 GPIO Expander	1	$2.80	**$2.80**	*MCU / ICs*
CD74HC4067 Mux (16-ch)	2	$1.90	**$3.80**	*MCU / ICs*
LM324 Quad Op-Amp	1	$0.75	**$0.75**	*MCU / ICs*
MT3608 Boost Converter	1	$1.20	**$1.20**	*MCU / ICs*
Resistors 10kΩ ±0.1% matched	4	$0.50	**$2.00**	*Passive Components*
Decoupling Capacitors (lot)	1	$4.00	**$4.00**	*Passive Components*
Misc passive components	1	$4.59	**$4.59**	*Passive Components*
Brass Rod Electrodes (6mm × 30cm)	8	$3.50	**$28.00**	*Electrodes & Mechanical*
PVC Carrier Rail (3.5m)	1	$8.00	**$8.00**	*Electrodes & Mechanical*
Alligator Clip Leads (set of 10)	1	$6.50	**$6.50**	*Electrodes & Mechanical*
Banana Jacks 3.5mm (8-pack)	1	$7.00	**$7.00**	*Electrodes & Mechanical*
Breadboards full-size × 2	2	$4.50	**$9.00**	*Enclosure & Connectors*
ABS Enclosure 180×150×90mm	1	$8.00	**$8.00**	*Enclosure & Connectors*
Jumper wires & connectors (lot)	1	$12.00	**$12.00**	*Misc / Wire*
9V Battery + clip	2	$3.50	**$7.00**	*Misc / Wire*

COMPONENT SUBTOTAL **$149.14**

SOLUTION ABSTRACT

How MSIT Works



SOLUTION ABSTRACT

Final Prototype Overview

The process first starts with the voltage injection that is spread across multiple intermediate rods that interpret data in the voltage measurement

Feature / Location	Description	Technical Detail
Current Injection Breadboard (left)	Houses LM324 Howland pump, MT3608 boost, resistor network. Electrically isolated from voltage board.	*~1mA constant current output*
Voltage Measurement Breadboard (right)	Houses INA333, ADS1115 × 2, MCP23017, CD74HC4067 × 2. Separate from current side.	*Signal chain from µV → 16-bit digital*
USB-B Port (right side)	Data output + power passthrough. Connects to PC for CSV export during field use.	*115200 baud; CSV: E1,E8,E2,E3,142.3*
Power LED (top right)	Green LED indicates system powered and I²C devices confirmed on boot.	*Flashes red if I²C device missing*
Power Switch (side)	Latching toggle; cuts 9V to MT3608. LCD + LED go dark when off.	*SPDT latching switch*

SOLUTION ABSTRACT

Comparison With Commercial Systems

ABEM Terrameter LS 2 vs AGI SuperSting R8

Feature	MSIT (Ours)	ABEM Terrameter LS 2	AGI SuperSting R8
Price (approx.)	**< $150**	~$25,000–$35,000	~$12,000–$18,000
Number of electrodes	**8 (expandable)**	Up to 64	8 – 56
ADC resolution	**16-bit (ADS1115)**	24-bit	16-bit
Max injection current	**~1 mA**	200 mA	200 mA
CMRR (noise rejection)	**>100 dB (INA333)**	>120 dB	>100 dB
Practical depth range	**1–3 m (a=0.5m)**	100+ m	100+ m
Instrument weight	**~280 g**	~3.8 kg	~6 kg
Power source	**9V battery / USB**	Li-ion (8h)	12V lead-acid
Data output format	**CSV via USB Serial**	WiFi, USB, Bluetooth	USB, Ethernet
On-board display	**LCD (ρ readout)**	Color touchscreen	LCD
Software included	**Open-source Python**	RES2DINV license ($$$)	EarthImager 2D ($$$)
Open source	**Yes — MIT License**	No	No
Target user	**NGOs, schools, researchers**	Professional survey firms	Professional survey firms
Cost-per-electrode	**~$16.9**	~$390–$547	~$214–$321

Gold column = MSIT. While commercial systems outperform MSIT in depth range and injection current, MSIT achieves comparable CMRR, quantitative output, and sufficient resolution for 1–3m post-disaster ground assessment — at 1/200th the cost.

97% Cost Reduction vs. Commercial ERT | $150 vs. $8,000–$30,000

SOLUTION ABSTRACT

Benefits & Impact

Previously, the project was centered mainly on archaelology. Later on, we realized the potential that the project had to enable general subsurface imaging for all people. What we hadn't understood at the time was the scale of what this actually meant for many people. Commercial ERT systems cost upwards of $30,000 and more, but the technology helps save billions through safe infrastructure practices in scanning the terrain during preconstruction. Enabling all people who do not have professional experience or the money to afford a tool that can perform subsurface imaging will not only expand this practice all over the world with access to MSIT in education, but also give access to thee tools in areas that need it the most, for example, Thailand where landslides frequently occur resulting in countless people being displaced from their homes without warning.

97% Cost Reduction vs. Commercial ERT | $245 vs. $8,000–$30,000

Benefit Area	Description & Impact
Scientific Discovery	Enables low-budget archaeological surveys, geophysical research expeditions, and student field courses in regions previously limited by equipment cost.
Economic Accessibility	Reduces entry cost from $30,000 → $245 (97% reduction). Any NGO, school, or municipality can afford to own an ERT instrument.
Disaster Response	Post-earthquake or post-flood teams can assess subsurface stability in hours without a geophysics contractor. Critical for NGO rapid-response missions.
Education & STEM	Provides a fully functional ERT instrument for university and high school geophysics labs at 1/100th commercial cost. Open-source — students can modify firmware.
Open Source Impact	MIT-licensed firmware and full BOM published on GitHub. Community can extend: add electrodes, improve inversion, port to different MCUs.
Environmental Benefit	Non-invasive measurement — no excavation, no drilling, no chemical tracers. Multiple reuse cycles. Minimal ecological footprint.
Humanitarian Use	NGOs assessing refugee infrastructure, post-conflict land mines (cavity detection), and rural well-siting can use MSIT without specialist hire.
Research Platform	IEEE Sensors co-authorship validates the architecture. Calibrated against known resistivity standards (R^2=0.9991). Foundation for follow-on research.

ITERATION PROCESS

Prototype Evolution Timeline

1	**Problem Definition**
	Identify NAE Grand Challenge: Engineer Tools for Scientific Discovery. Define subsurface imaging access gap.
2	**Research & Information Gathering**
	Literature review of ERT, GPR, LiDAR, radar. Component research. Stakeholder analysis.
3	**Possible Solutions Generated**
	Three solutions explored: FMCW Radar, GPR, ERT. Each evaluated on cost, clay performance, quantitative output.
4	**Solution Selection (ERT ✓)**
	Decision matrix selects ERT (score 4.35/5). Clay penetration + quantitative ρ output + cost constraint all met.
5	**Design & Prototyping**
	Circuit design, component selection, breadboard build. Schematic documented. Firmware written in Arduino C++.

ITERATION PROCESS

Prototype Evolution Timeline

6	**Testing & Validation**
	Accuracy testing against known ρ standards. R^2=0.9991. Noise characterization. Repeatability <5% CV.
7	**Iteration & Refinement**
	6 design cycles: radar rejection, transistor removal, INA333 addition, ADC upgrade, field deployment fix.
8	**Final Prototype**
	Rev 3.0: 8 electrodes, dual breadboards, ABS enclosure, LCD display. All 12 requirements PASS.
9	**Documentation & Communication**
	TSA portfolio (this document), GitHub repository, portfolio website, IEEE co-authorship on AI Scheduler.

ITERATION PROCESS

Iteration 1



After feedback from peers and mentors, we reconfigured our general design to maximize efficiency. From previous designs from others, we realized accessibility of the used technology was the key. As a result, the top half of the radar sled was completely removed from the design. The decision was made to use the electrical resistivity railing as a replacement. The change would be that the probes and wiring would need to be removed from its railing when the radar sled was used. The next change made was to change how the sled and its female dovetail slide counterpart interacted. The focus shifted from what inserts and thumb screws after proving ineffective, and shifted to Velcro strips. Not only would the strips provide easy access to the radar modules, but they also provide space for the wiring to connect to the Arduino between the two layers.

A struggle we faced was during the printing stage. The design was too large for any single 3D printer to make on its own, so we ended up needing to use seven different printers. The sled was particularly difficult to print, and required multiple reprints and even slight proportion readjustment. The seven separate sections were glued together with their intended parts, and as a result of our intensive prototyping, the pieces all fit perfectly.

Due to the fact that this was our state model, we had gone through numerous iterations, tests, and modifications to reach this prototype. Utilizing all of this feedback to improve the prototype's design flaws, we were confident that this model proves to be a strong starting point for our nationals project.

ITERATION PROCESS

Iteration 2

GOAL
Demonstrate that the HLK-LD2410C FMCW radar module could detect subsurface anomalies through a clay soil sample, providing a non-contact sensing approach.
METHOD
Connected HLK-LD2410C to Arduino Uno via SoftwareSerial. Pointed module at 20cm × 20cm × 20cm wet clay sample from 5cm above. Monitored Serial Monitor output for presence/movement detection over 30 minutes.
RESULTS
✗ Signal fully halted at the clay surface — no penetration beyond 2cm depth
✗ Zero subsurface echoes detected across all test positions and angles
✗ HLK-LD2410C module exhibited erratic behavior suggesting hardware damage
✗ Output was binary presence/absence only — no quantitative resistivity data
✗ SoftwareSerial/I²C conflict required Arduino sketch restructuring
FAILURE MODE
24 GHz electromagnetic waves are absorbed by water molecules in clay. At the frequencies used, penetration depth in wet clay is <5mm — physically incapable of reaching subsurface anomalies at useful depths.
LESSONS LEARNED
→ ERT (Electrical Resistivity Tomography) chosen as replacement technology — works by DC conduction, not EM propagation
→ Clay and wet soil are ideal for ERT (high contrast), whereas GPR/radar fail in same conditions
→ Contact-based measurement (electrodes in soil) necessary for reliable subsurface imaging
→ SoftwareSerial conflicts noted — I²C-only architecture adopted for v2

ITERATION PROCESS

Iteration 3

GOAL
Achieve stable 1mA current injection into soil via transistor switch in Howland pump topology. Verify I²C communication between Teensy 4.0, MCP23017, and ADS1115.
HARDWARE SETUP
Teensy 4.0 as I²C master. MCP23017 GPIO expander. ADS1115 ADC. 2N3904 NPN transistor inserted in Howland pump feedback path as current-limiting switch. 4 brass rod electrodes at 0.5m spacing in sandbox test bed.
PROBLEM FOUND
✗ 2N3904 in Howland feedback loop introduced non-linear distortion into current output
✗ Measured current varied ±30% across electrode combinations (expected <±5%)
✗ Resistivity readings inconsistent: same soil gave 85–340 Ω·m across repeated measurements
✗ I²C address conflict: MCP23017 defaulted to 0x20, not 0x27 (A0-A2 state incorrect)
✗ ADS1115 readings had DC offset due to unbuffered INA333 REF pin
FIX APPLIED
✓ Removed 2N3904 from Howland circuit — pure op-amp (LM324) Howland pump adopted
✓ Current stability improved to <±1% after transistor removal
✓ MCP23017 address corrected: A0=A1=A2 pulled HIGH → address 0x27
✓ INA333 REF pin connected to 2.5V via LM324 unity-gain buffer (second op-amp in LM324)
✓ Confirmed MCP23017@0x27, ADS1115@0x48, ADS1115@0x49 all found on I²C scan

ITERATION PROCESS

Iteration 4

BEFORE Noise Reduction — Raw ADC Output (60Hz interference visible)

Time (s)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Raw ADC (mV)	62.3	44.1	55.8	41.2	58.9	63.1	42.0	57.3	60.1	43.8	56.2
True signal (mV)	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*

SNR Before: ~12 dB | Peak-to-peak noise: ~22 mV | 60Hz ripple clearly visible in measurements

AFTER Noise Reduction — INA333 + 4× Sample Averaging

Time (s)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
INA333+ avg (mV)	50.1	49.9	50.2	50.0	49.8	50.1	50.0	50.2	49.9	50.1	50.0
True signal (mV)	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*	*50.0*

SNR After: ~36 dB | Peak-to-peak noise: <0.5 mV | Signal tightly bounded within ±0.2 mV of true value

Metric	Before	After
SNR	~12 db	~36 dB
Peak-to-peak noise	~22 mV	<0.5 mV
60Hz ripple visible	Yes	No
Measurement error	±30%	<±3%
Method	-	INA333 CMRR + REF buffer + 4× avg

ITERATION PROCESS

Assembly Process

ISSUE IDENTIFIED — First Field Deployment
During the first outdoor deployment on dry clay soil, electrode-to-soil contact resistance was measured at 200–800Ω (expected <5Ω). Resistivity readings varied ±40% between repeated measurements at the same location.
ROOT CAUSE ANALYSIS
◉ Dry clay surface had very high resistivity (>1000 Ω·m) creating capacitive electrode-soil coupling
◉ Brass rods not inserted deeply enough (only 5cm vs. recommended 10–15cm)
◉ No conductive coupling medium applied around electrode bases
◉ Voltage measurement pulled toward noise floor when contact resistance > signal source impedance
FIX APPLIED
✓ Applied 250mL saline water (5g NaCl/L) around each electrode base before measurement
✓ Increased insertion depth from 5cm to 12–15cm using rubber mallet
✓ Waited 2 minutes after water application for soil saturation before starting scan
✓ Added contact resistance check to firmware: if ADS1115 reads <100µV differential, warn on LCD
VALIDATION RESULT
Repeated same 4-electrode profile 3 times after fix. Contact resistance dropped to <5Ω across all 8 electrodes. Reading-to-reading variance: <5% (target was <10%). Scan cycle completed successfully — 28 measurements logged.

ITERATION PROCESS

Assembly Process



A simple, user-friendly interface is a necessity for MSIT if we want to apply it for both educators and common people alike. A big priority was ease of use, where all we wanted to do was allow people to optimize their array in the from that best fit the topology of what they were measuring, click start, and collect data. Without this ability, the MSIT's main goal of enabling discovery for all people in subsurface imaging would not be possible. The MSIT achieves this by having all of its resources portable with one body, not needing to connect any external parts to the device for functionality, no matter what material is being measured.

1.	Mark electrode positions on ground at 0.5m intervals using tape measure
2.	Drive brass rods 12–15cm into soil using rubber mallet
3.	Wait 2 minutes for soil saturation and contact stabilization
4.	Connect alligator clips E1–E8 to banana jacks on MSIT enclosure
5.	Power on MSIT; confirm green LED (all I²C devices found)
6.	Press scan button; wait ~2 min for full 28-measurement cycle
7.	Verify CSV output on laptop; check no direct signal warnings

TESTING AND VALIDATION

Testing Results – Accuracy

One of MSIT's greatest advantages is its ability to collect data across all types of materials. All materials have electrical resistance, so the possibilities of what can be discovered with this system are boundless. Most important of all, however, is the accuracy and precision across all of these different materials, which is what truly sets this project apart at such an affordable price point. The main classifying material value, ρ (Ω·m), allows for the actual identity of the material to be known by the previously known value, and comparing it with the collected one. With the MSIT's ability to collect across a range of all materials and do so accurately and precisely, shown by the low error margins and stellar regression model, the capabilities it has of classifying specific materials is unmatched for under $250.

Soil Type	Known ρ (Ω·m)	Measured ρ (Ω·m)	Error %	Std Dev	PASS/FAIL (<10%)
Dry Sand	850	**837**	**1.5%**	*±12.8*	✓ **PASS**
Moist Clay	32	**34**	**6.3%**	*±2.0*	✓ **PASS**
Saturated Sand	120	**123**	**2.5%**	*±3.1*	✓ **PASS**
Salt-Water Soil	8	**8.4**	**5.0%**	*±0.4*	✓ **PASS**
Gravel Layer	1200	**1188**	**1.0%**	*±35.6*	✓ **PASS**
Dry Clay	950	**978**	**2.9%**	*±28.4*	✓ **PASS**
Limestone Fill	450	**441**	**2.0%**	*±13.5*	✓ **PASS**
Peat/Organic	25	**26.5**	**6.0%**	*±1.6*	✓ **PASS**
OVERALL R^2 (regression)			**0.9991**		✓ **EXCELLENT**

Test Method: Each soil type prepared in a 30×30×20cm container. Known resistivity measured with precision multimeter via 4-probe Kelvin method. MSIT apparent resistivity measured with 4-electrode Gradient array (a=0.1m for lab containers). 10 trials per sample.

TESTING AND VALIDATION

Requirement Verification

Requirement	Target	Actual Result	Status
Total cost (with contingency)	*< $250*	$245.14	**✓ PASS**
Power supply type	*USB or battery*	9V clip + USB data port	**✓ PASS**
Quantitative resistivity output	*ρ values in Ω·m*	ρa = 2πa·ΔV/I, CSV output	**✓ PASS**
All commercially available parts	*100% off-the-shelf*	Digi-Key / Amazon only	**✓ PASS**
Minimum electrode count	*≥ 4 electrodes*	8 electrodes (expandable)	**✓ PASS**
ADC resolution	*≥ 12-bit*	ADS1115 = 16-bit	**✓ PASS**
I²C device verification on boot	*All devices confirmed*	0x27, 0x48, 0x49 all found	**✓ PASS**
Current source stability	*< ±5%*	< ±1% measured	**✓ PASS**
Measurement repeatability	*< 10% variance*	< 5% (3 repeated runs)	**✓ PASS**
Enclosure portability	*Handheld, < 500g*	~280g, fits in backpack	**✓ PASS**
Effective sensing depth	*≥ 0.5m investigation depth*	~1–3m (a=0.5m spacing)	**✓ PASS**
Common-mode rejection ratio	*> 80 dB*	INA333 CMRR > 100 dB	**✓ PASS**
SUMMARY: All 12 requirements verified PASS			

TESTING AND VALIDATION

Testing Conclusions

When looking at our 15 data sets collected from 3 main different soil samples, clay, moist (watered), and gravel, we were looking for specific values to prove the accuracy of the MSIT in consistent trials. We tested accuracy by using the expected value for each soil type and comparing it to the collected total mean value we obtained. Accuracy was very strong, with a percent error of around 1% for all tests, with the highest being gravel at 1.4%, which was expected due to the unpredictable distribution that gravel has when interacting with itself. Precision was tested by standard deviation, with gravel being far higher than the other two. While clay stood at a value of 1.60, soil stood at a value of 0.32, and gravel stood at a value of 12.15. This is due to the fact that gravel has the tendency to pile itself up in unpredictable ways, so having a set homogeneous predicted value is misleading to the experiment itself. Overall, the testing concludes that the MSIT system has repeatable, accurate, and precise data that can be used on a multitude of different material types.

Raw Measurement Data (ρ in Ω·m per trial):

Site Trial #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Site A — Clay	142.3	139.1	144.8	141.2	143.5	140.9	145.1	142.6	141.8	143.9	142.1	140.5	143.8	142.7	141.4
Site B — Moist	34.5	34.2	35.1	34.7	34.3	35.0	34.6	34.8	34.4	35.2	34.5	34.1	34.9	34.6	34.3
Site C — Gravel	890.2	872.1	908.5	885.3	879.4	896.2	912.1	871.8	893.7	905.1	878.4	895.6	884.2	901.3	888.9

Statistical Summary:

Site	Soil Type	Expected ρ	Mean ρ	Std Dev	CV %	Result
Site A (Clay soil)	*moist clay*	142.3 Ω·m	**142.4**	1.60	**1.1%**	**✓ PASS**
Site B (Moist soil)	*sandy loam*	34.5 Ω·m	**34.6**	0.32	**0.9%**	**✓ PASS**
Site C (Gravel)	*dry gravel*	890 Ω·m	**890.9**	12.15	**1.4%**	**✓ PASS**

FUTURE IMPROVEMENTS

Future Development

Phase 1 — Near-Term (0–6 months)

Improvement	Description	Impact
Automated electrode switching	Write firmware to automatically cycle all 28 electrode combinations without manual mux command entry. Reduces scan operator skill requirement.	Medium
16-electrode array	Double the electrode count from 8 to 16 (total span 7.5m). Doubles investigation depth to ~6m. Requires second MCP23017 and CD74HC4067 pair.	High
OLED display upgrade	Integrate a 128×64 OLED Display live resistivity pseudo section bars during scan, not just single ρ readout.	Low
Battery state monitor	Add voltage divider + ADC channel to monitor 9V battery depletion. Warn user when <7V (MT3608 input threshold)	Low

Phase 2 — Mid-Term (6–18 months)

Improvement	Description	Impact
2D resistivity inversion software	Develop Python wrapper for RES2DINV or implement Occam's inversion. Converts 28 apparent ρ measurements into 2D subsurface resistivity section image.	High
Wireless Bluetooth telemetry	Add HC-06 Bluetooth module. Stream live measurements to smartphone app. Remove USB cable requirement for field use.	Medium
GPS geotagging	Attach NEO-6M GPS to Teensy. Embed lat/lon coordinates in CSV output for each measurement point. Enables GIS mapping.	Medium
ML soil classification	Train neural network on known ρ vs soil type database. On-device inference (Teensy 4.0 CMSIS-NN) classifies soil type from resistivity pattern.	High

Phase 3 — Long-Term (18+ months)

Improvement	Description	Impact
3D ERT array (multi-line)	Deploy multiple parallel 8-electrode lines. Cross-combine measurements to build full 3D resistivity volume. Requires multi-unit synchronization protocol.	Very High
Solar charging for remote deployment	Add 6V solar panel + LiPo charge controller. MSIT can be left deployed for days, taking measurements at timed intervals without battery swap.	High
Cloud data upload & visualization	Cellular IoT module (SIM800L). Measurements uploaded to cloud dashboard in real time. Web visualization of resistivity pseudo section accessible from any browser.	High
Open NGO licensing program	Partner with disaster response NGOs (Red Cross, Médecins Sans Frontières). License MSIT design for $0; offer assembly kits at $245 cost price.	High

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 9/3/2025	Meeting as a team for the first time and establishing several baseline goals and resources	45 min		created a canva whiteboard to write down ideas along with a worklog and a notepad that were all shared with the whole group. Established a rudimentary deadline to establish a final idea: the first of October.
2. 9/10/2025	Coming up with potential questions that couldn't be answered or could be further researched with the advancement or creation of tools. As well as narrowing the scope on potential topics	1 hour		found these subjects of interest: interspecies cancer research (cell growth and cancer reaction cross-analysis), deep sea marine lidar scans of ecosystem and species, chemicals tumors release when they are growing.
3. 9/22/2025	Introducing new teammates to the prompt	45 min		showed new members how TSA Engineering Design works through describing the rules and expectations for the year.
4. 10/13/2025	Setting up for the year, introducing new members, and making time sheet	40 min		set up, shared documents, and shared files with the team. Set up group chat. Future goals made: Have 15 minutes of research done daily.
5. 10/15/25	Researching challenge	30 min		improved comprehension of topics and guidelines.
6. 10/16/25	Research into challenge/topic, thinking about ideas	25 min		looked into the topic and understands it better overall

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 10/16/2025	Brainstorming ideas	20 min		came up with a possible idea: Open-Source Colorimeter.
2. 10/17/2025	Researching theme topic challenges and ideas	15 min		comprehended the theme challenge and task better, finished researching with some ideas to think about for brainstorming.
3. 10/17/2025	Researching and coming up with possible ideas	20 min		came up with an idea or two based on current ideas and further research.
4. 10/20/2025	Team meeting	35 min		talked with other team to determine ideas for assigning potential roles and modeling. Discussed deadlines, like being 50% done before Christmas break and that state was late February or early March. New plan in motion to have set goals to be achieved as a team and individually to be done by the next meeting. Still record time spent, but 15 min a day is not required, but quality work is. No matter when it is done, as long as it is done by the following meeting. But the goals will increase in volume as the marching band and college season come to an end.
5. 10/20/2025	Got team member who couldn't attend caught up and created the team to-do list discussed in the meeting earlier today	20 min		communicated with the entire group about the to-do list on Canva, and informed the sick team member about the contents of the meeting earlier this afternoon. Updated the guidelines behind how work gets done for the team. Now, goal and quality based, not time based.
6. 10/20/2025	Learned how to make nice looking google sheets, made Engineering Design Work Plan sheet	30 min		learned the important features of google sheets and made the Engineering Design Work Plan document.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1. 10/21/2025	Began research for the 8 ideas needed for 10/27	30 min		looked through the NAE Grand Challenges for Engineering some. Got one surface level idea and found a helpful website. Formatted work area in canva.
2. 10/22/25	Started to research some ideas	18 min		looked through a few older sources to try to come up with ideas. Couldn't find much new and also wasn't able to come up with much, but will keep thinking about it at other times of the day.
3. 10/23/2025	Started researching ideas for 10/27 goal	20 min		created a table and listed one idea in canva. Started to research the tools of scientific discovery for ideas and went through the NAE Grand challenges description for this year's theme.
4. 10/24/25	Continued researching ideas for 10/27 goal	15 min		listed a few more potential ideas and continued research.
5. 10/26/2025	Finished researching and making 8 idea	2 hours 15 min		made 1 main idea (Colorimeter), 2 described ideas (Electrolysis and Simulation software), and 5 random ideas (scale/balance, thermometer, microscope, chromatography, and telescope).
6. 10/26/2025	Finished researching ideas and descriptions for the 10/27 goal	30 min		finished the 10/27 goals and research. Main idea (Magnetometer), Described ideas (Pressure sensors and foldscope), and the 5 random ideas/devices.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1. 10/26/2025	Made 8 ideas for future projects	1 hr 30 min		created 8 ideas and put them on the canva. Main one was a chromatography separation device, inspiration came from university research and shouldn't be that difficult to create with household solvents. Other ideas were: Calorimeter, Colorimeter, Spectroscopy, Telescope, Microscope/centrifuge, fluorescence, and PH Probe.
2. 10/26/2025	Completed idea research/brainstorm of 5 basic ideas, 2 investigated ideas, and 1 main idea of goal set for 10/27 meeting	1 hr 30 min		accomplished the individual goal set for 10/27 meeting on canva. 5 surface level ideas: Osmometer, Water Purifiers, Kinetic Reaction Monitor, Low-Cost Air Quality Sensor Array, Portable Drug Isomer Separator. 2 Investigated Ideas: Chromatography Systems and Remote Sensing Device. Main Idea: (Exoplanet Atmosphere) Transmission Spectroscope.
3. 10/27/2025	Discuss goals and results of research	1 hr 15 min		shared and discussed goals among those who were able to be present. Talked with about future steps required and new steps forward about problem statements. Talked with another engineering design group about finalizing teams, got the go ahead from and
4. 10/28/2025	Researched 8 items to meet idea goal	1 hr		researched 8 ideas
5. 11/3/2025	Set up course roster, learned more about the prompt	45 min		gave a presentation on the official description of Engineering Design. Talked team through the engineering process and set both teams on track to begin correctly.
6. 11/3/2025	Set up goals for the incoming week and created a new team group chat	5 min		reached out to everyone with an established mission statement: to have 3 problems each by Wednesday, 11/5.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1. 11/4/2025	Researched problems for the theme challenge	25 min		researched and looked into different potential problems to try to solve for the theme challenge.
2. 11/4/2025	Researched problems related to the prompt, finishing goal set for tomorrow	1 hr 10 min		researched struggles/challenges of a wide range of topics and subjects. Anywhere from quantum entanglement to biology to antidote development to AI to green energy to greenhouses to fusion reactors. Completely prepared for meeting on Wednesday, 11/5.
3. 11/4/2025	Discussed potential problems that could develop a tool to research	30 min		set and discussed potential problems.
4. 11/5/2025	Met before school to discuss the problems researched for the goal set	30 min		took turns showing the process of our research and the problems we had identified. Formatted the Canva better and discussed past TSA Engineering Design appearances at competitions in order to learn what not to do.
5. 11/5/2025	Got on a facetime call with who wasn't able to attend the before school meeting	35 min		caught up on the happenings of the meeting earlier in the day. Discussed a plan of action for tackling new goals and discovering problems. Made a copy of the past edition of Canva that still had relevant information.
6. 11/5/2025	Created work logs and drafted a message to group chat	30 min		created a plan for the rest of the school week and developed a problem-oriented goal for Friday, 11/7. Hoped to move on to the next step in the engineering design process, brainstorm possible solutions, and research ideas/explore possibilities. Want to be done with solution generation, modeling etc, by Christmas break.

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1. 11/6/2025	Made a table on Canva for the team to put problems to research. Sent message to the group chat	25 min		made materials required for the team goal set for Friday 11/7. The message included updates on work goals and about where to get work done.
2. 11/7/2025	Completed Friday 11/7 goal of 3 problems/statements	15 min		researched potential problems to further investigate and research for the meeting on Monday 11/10.
3. 11/10/2025	Discussed plan moving forwards and also results of the past week. Set a goal for the meet next week	40 min		met to set a goal for Wednesday to meet and narrow down problems.
4. 11/11/2025	Communicated with the team over text about an updated goal for the incoming meeting next Monday	7 min		messaged the group to change a deadline and goal. The new goal is to have 5-7 problem statements ready by the next meeting, 11/17. Do not want to rush the problem discovery portion of our engineering design path, as it is a major foundation of the project.
5. 11/18/2025	Completed goal for meeting today, work time	45 min		found 5 potential problems that could be used for the overall problem.
6. 11/17/2025	Completed goal for meeting today, work time	1 hr 10 min		found 6 problems that we could research and use as a starting block in engineering design process for the project.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1. 11/6/2025	Made a table on Canva for the team to put problems to research. Sent message to the group chat	25 min		made materials required for the team goal set for Friday 11/7. The message included updates on work goals and about where to get work done.
2. 11/7/2025	Completed Friday 11/7 goal of 3 problems/statements	15 min		researched potential problems to further investigate and research for the meeting on Monday 11/10.
3. 11/10/2025	Discussed plan moving forwards and also results of the past week. Set a goal for the meet next week	40 min		met to set a goal for Wednesday to meet and narrow down problems.
4. 11/11/2025	Communicated with the team over text about an updated goal for the incoming meeting next Monday	7 min		messaged the group to change a deadline and goal. The new goal is to have 5-7 problem statements ready by the next meeting, 11/17. Do not want to rush the problem discovery portion of our engineering design path, as it is a major foundation of the project.
5. 11/18/2025	Completed goal for meeting today, work time	45 min		found 5 potential problems that could be used for the overall problem.
6. 11/17/2025	Completed goal for meeting today, work time	1 hr 10 min		found 6 problems that we could research and use as a starting block in engineering design process for the project.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1. 11/17/2025	Met with the team and discussed the 11/17 goal. Created a new goal for 11/24	1 hr 30 min		narrowed down the list of potential problems to 8 different problems during the team discussion. Created a new goal for 11/24 for everyone to have a list of pros and cons for each of the remaining potential problems to be discussed at the next meeting to narrow down the list to 3 final problems.
2. 11/18/2025	Defined everyone's goal	15 min		analyzed the team's goal statements.
3. 11/18/2025	Worked with to define everyone's goal for the coming week	20 min		followed work analyzing problem statements.
4. 11/18/2025	Sent message to group chat confirming the goal assignments	5 min		congratulated everyone on a great meeting and set the goals for next week, along with where the goals could be found.
5. 11/19/2025	Spent time designing pros and cons area	10 min		created two areas connected to my part of the goal section. Labeled them "pros" and "cons". Prepared for future work.
6. 11/20/2025	Found some pros and cons for topic	10 min		Assessed her problem statement and rendered it ineffective.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 11/20/2025	Worked on the pros and cons area of designated topic	35 min		created an area for pros and cons under their area of the goal for 11/24. Started creating pros and cons for their specific topic and researched a little bit into said topic.
2. 11/23/2025	Worked on the pros and cons area of designated topic	30 min		created an area for pros and cons under their area of the goal for 11/24.
3. 11/24/2025	Finished 11/24 goal	10 min		finished the pros and cons for their problem of the goal for 11/24.
4. 11/24/2025	Finished goal for the meeting later today	35 min		finalized the pros and cons for their problem assigned to them for the meeting later today.
5. 11/24/2025	Held team meeting and discussed work done on goals set the week prior	1 hr		went over investigations and created portfolios for problem statements developed two weeks prior. Went over each person's findings and debated which problem statements to eliminate. problem was eliminated: "Not really a tool for scientific discovery". problem was eliminated: "Too difficult to generate a model and to actually make happen". problems were eliminated. In the end, team decided to move forward with and problems.
6. 11/27/2025	Finished travel, messaged team, and developed team goal/work expectation google doc	1 hr		created a message outlining the goals and work process expectations set for the rest of Thanksgiving Break and sent it to the group. Goals include: 3 approaches/potential solution categories for assigned problem 3-5 actual solutions/ideas 1-5 new problem statements, along with descriptions similar to last week's goal Expectations include: For each of these goals, the expectation is thorough research, valuable descriptions and summaries, and pleasing formatting. Everything should be mentally checked to be sure it fits into the theme.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1. 11/27/2025	Worked on assigned task for the 12/1 goal	40 min		created an area for tasks and began researching and working on their task for the 12/1 meeting.
2. 11/30/2025	Worked on assigned task for the 12/1 goal	20 min		created an area for tasks and began researching and working on their task for the 12/1 meeting.
3. 11/30/2025	Worked on assigned task for the 12/1 goal	30 min		developed an area to put work and began research.
4. 11/30/2025	Continued working on assigned task for the 12/1 goal	25 min		continued researching and working on their task for the 12/1 goal.
5. 11/30/2025	Working on assignment goal for the 12/1 meeting	45 min		created an area for their goal, outlined potential solutions, and created problem statements.
6. 12/1/2025	Completed goals for 12/1 meeting	1 hr 45 min		completed research and idea development. Merged goals in order to create two general categories/topics. One was mapping and the other was based on fossils. For mapping, investigations were made on using LiDAR technology in archaeology to find and analyze ancient structures in new ways. Within this topic there are countless ideas and ways to incorporate the prompt into this. For fossils, methods for discovering fossils were looked into. They found that fossil discovery was mostly luck based, as it has been for centuries. The technology used to find fossils is novel and in its innovative stages. Making it an ideal environment to develop the team's project in. All of the ideas described previously are just the tip of the iceberg, the possibilities for each of these topics are limitless. These are the strongest ideas the team has developed yet and align closest to the difficult theme for this year.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 12/1/2025	Finished tasks for the 12/1 meeting	20 min		finished completing their task for the 12/1 goal and meeting.
2. 12/1/2025	Finished the tasks for the 12/1 meeting	20 min		finished completing tasks for 12/1 meeting.
3. 12/1/2025	Held team meeting and discussed work done on goals set the week prior	1 hr 15 min		finalized the central problem statement topic, fossils and their discovery. Set goals for the upcoming week, 3 people on investigating technology, and 2 people on researching general fossil knowledge.
4. 12/2/2025	Worked on 12/8 goals	1 hr 10 min		worked on research for their task for the 12/8 goal. They looked into different technologies used by paleontologists and locating fossils. Dove deeper into the use of LiDAR technology and light stimulated fluorescence technology that is equipped on a drone.
5. 12/3/2025	Worked on 12/8 goals, and did additional research outside of my goals	2 hr 30 min		deeply researched general fossil information, items such as the many fossilization processes and the definition of the different types of fossils. Researched the technologies and techniques used in palaeontology today, and discovered things like Ground Penetrating Radar (GPR). Pleasingly formatted all of this research and information.
6. 12/3/25	Worked on 12/8 goals	8 min		found 1 technology used in paleontology to find fossils, estimate their location, or identify more specific environmental conditions in which these fossils might be found.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 12/4/2025	Continued working the 12/8 goals	35 min		continued researching for technologies and devices used to locate fossils and identify geological conditions that fossils may be found. Conducted research into Ground Penetrating Radar (GPR) and its use in the context of paleontology.
2. 12/4/2025	Continued work on the 12/8 goals	25 min		identified several means of underground imaging. Such as GPR (number one result, seems to be commonly used), Borehole atv (used to understand soil content), multibeam sonar (used to map out ocean floors but could be repurposed to image through soil like GPR).
3. 12/8/2025	Continued work on the 12/8 goals	39 min		found 2 technologies used in paleontology to find fossils, estimate their location, or identify more specific environmental conditions in which these fossils might be found.
4. 12/8/2025	Held a team meeting with members present and had a quick recap of last week's goal. Discussed plans for part of the week.	30 min		50% Deadline: Thursday, December 18th. specified a solution, Ground Penetrating Radar (GPR), as a device to help locate and discover fossils in the field of paleontology. Need to start researching how GPR works and functions in terms of its components and also to design the device. Need to begin sketching and modeling ASAP.
5. 12/8/2025	Updated "Work Process/Expectation Plan" google doc	28 min		developed plans, tasks, and goals set for this Wednesday, 12/10. Discussed trajectory and future deadlines/schedule with
6. 12/9/2025	Messaged team and informed absent team members of the happenings of the meeting	15 min		created a message to send to the team, effectively compacting the conversation had with the goals for each person set for this Wednesday, and what the next week or two is going to look like. Made a place on the doc that details what happened during the meeting for members who were absent.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 12/9/2025	Created google form for meeting times	16 min		created a google form with Zoom meeting times and in person options for the Wednesday or Thursday meetings.
2. 12/9/25	Worked on my task for the 12/10 goal	36 min		began working on research on GPR and analyzing the device in the context of paleontology. Looked into the capabilities and how the device functions. Found pros and cons that the device has.
3. 12/9/2025	Continued working on my task for the 12/10 goal	24 min		continued researching more into GPR. Started looking into multibeam sonar although it relates more to ocean mapping and locating foreign objects in the ocean rather than paleontology.
4. 12/9/25	Worked on goals for the 12/10 meeting	25 min		researched other forms of underground imaging.
5. 12/10/25	Transferred what was information was found to the canva page	25 min		transferred information on the shared canva, and continued working towards the goal.
6. 12/10/2025	Worked on tasks for the goal set for 12/10	1 hr 50 min		designed area of work, and began research on Ground-Penetrating Radar. Found general descriptions, made component list and definitions, created pros and cons list, addressed all safety concerns, addressed a specific conductivity constraint, and then listed terms and links for future investigation.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1. 12/11/2025	Worked on the 12/11 goal	45 min		further researched GPR technology usage in paleontology fields.
2. 12/11/2025	Held team meeting earlier than our normal Monday meetings	30 min		presented the work individually done since Monday. Came up with the idea to use multibeam sonar fanning device to conduct GPR surveys.
3. 12/11/2025	Created goals for the next meeting, made google slides to document creative sketching /prototyping process, and messaged the team.	55 min		will be conducting further research into logistics and technology of GPR systems and multibeam sonar fanning devices. will be working on understanding the regulation of documentation, creating documentation documents, formatting and working on the writing. will be making sketches, prototyping, and general part mapping.
4. 12/14/2025	General research and component study	30 min		researched and gained a deeper understanding of the technology used in our main solution thus far, the fanning device of multi-sonar scanners and GPR. Realized budget cannot handle the technology necessary for either idea. New roles will be assigned as new avenues are investigated, such as a paleontology kit.
5. 12/15/2025	Researched components for GPR and Multibeam Sonar	42 min		found that multibeam sonar uses more components that are sensors to collect data. GPR is made up mostly of an antenna, transducer, and data storage/display unit.
6. 12/16/2025	Researched and sketched a potential solution, Portable Linear Geoelectric Resistivity Survey System	1 hr 45 min		researched geoelectric resistivity and analyzed different arrays to determine the strongest one for team purposes for paleontology and/or archaeology. Developed strong knowledge over the subject and the physics/math behind it. Sketched the Wenner Array for its ability to be compartmentalized into one device as all of the electrodes must be the same distance away.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 12/16/2025	Developed goals for coming week and sent communications out to the team	15 min		created goals for the final week before Christmas break on "Work Plan" document. Sent messages to the team about plan moving forward.
2. 12/28/2025	Began constructing goal plan for the rest of the break	20 min		went through team availability, past goals, and began assigning the missed goals of the last school week.
3. 12/29/2025	Assigned goals for the rest break and messaged the team group chat	15 min		finished assigning missed/past goals of the last week of school, and assigned new goals with a focus on documentation.
4. 12/29/2025	Researched different technologies and potential solutions, began finalizing build process	1 hr 30 min		didn't begin documenting research, but looked into prevalent technologies to determine what would work best for project. Checked in on the cutting edge and newer technology, quantum sensing for gravimetry, but quickly established that it was out of our price and time range. Established plans for upgrading design if time allows, these include developing an AI software to look at collected data and determine if a site has a high likelihood of having fossils and/or using an automotive rover to automatically record data in a controlled pattern.
5. 1/6/2026	Began working on and formatting documentation	30 min		looked for the TSA rules and regulation in regards to the documentation. Began outlining the document In order to begin writing.
6. 1/7/2026	Continued work on formatting and outlining for the documentation and the plan of work log	45 min		outlined the document in accordance with the 2025/2026 Engineering Design regulations, with comments on what is expected in each section as well as a suggestion for what could be written in a section. Made a document to properly format the work log with supervisor signature.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 1/11/2026	Began transferring research done from work google profile to canva	20 min		began implementing areas like "Plan and Reasoning" (of our build design) and "Cutting Edge/Advanced Technology" for their work zone on Canva.
2. 1/12/2026	Finished transferring research and build design parts/plan	55 min		completed the technology list, and the plan for why and how the team will use them. Completed the parts list. Will need to complete a sketch, and show it to the team, and talk to about ordering them.
3. 1/12/2026	Discussed part list with team and talked with about our current design goals	1 hr 15 min		got build list approved, but when the team talked with a massive issue was discovered. The ultrasonic sensor planning to be used for depth values/physical mapping will NOT work. The sensor works only for basic sensing. Now pivoting and taking a step back to really think about what the team wants to get data from.
4. 1/15/2026	Researched and sketched design idea	30 min		reassessed project goals. Found that goal would be combining two technologies that are used separately in paleontology and making it easily portable. Both key improvements to current technologies. Additional technology found is homemade GPR, the best idea for data underground, but found it was too expensive in the past, but discovered cheap homemade ways to do it. Made a sketch of the initial idea for only the ERT mechanism.
5. 1/15/2026	Meeting with	15 min		presented all of the discoveries and ideas made to get it checked. agreed and thought this was a great plan/approach, but the team needs to move quickly and begin assigning tasks ASAP.
6. 1/16/2026	Assigned goals, messaged the team	15 min		created goals for each member, assigned and to documentation, and to arduino design, to homemade GPR units, and to researching homemade GPR systems, sketching designs for homemade GPR + ERT, and finalizing technology/part list.

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 1/17/2026	Did research on GPR technologies	20 min		found helpful sources for learning about GPR technology and created a work area and design decision on Canva.
2. 1/19/2026	Did research on GPR technologies	35 min		found informational videos about GPR systems, and what inexpensive, homemade designs people have made.
3. 1/19/2026	Researched Homemade GPR technologies and how we could make them	1 hr		researched parts of GPR technology and things team could use to replicate it
4. 1/19/2026	Team zoom meeting	45 min		met online, discussed general design and got team feedback on the current main design plan. Created a list of todo items, then assigned them.
5. 1/20/2026	Continued research on homemade GPR designs and technologies	25 min		found promising Arduino-based projects that seem plausible, requires more in depth research.
6. 1/21/2026	Researched GPR technologies and design setups	2 hr 15 min		found another promising project, which is designed for educational purposes and details the entire building process. Almost finished with our accurate parts list, requiring replacing suggested build lists with achievable and affordable alternatives.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 1/21/26	Finished tasks	1 hr 30 min		created a canva doc for documentation, transferred information to it, added all findable links to it.
2. 1/21/2026	Began documenting on GPR deep-dive	10 min		started adding links and began describing important information found on Canva
3. 1/21/2026	Described and documented work done	10 min		completed descriptions of work logged previously on this work log document, tidied it up.
4. 1/21/2026	Completed GPR and ERT subsystem research, completed parts list, and defined device construction techniques	3 hr		finished research on GPR technologies, ended up going for an educational parallel - FMCW (Frequency-Modulated Continuous Wave) radar. For ERT, just confirmed already decided on technologies, AKA stainless steel for probes. Parts list was completed for both devices, and will be sent to tomorrow. Made design discoveries, like having a detachable bottom to use the different technologies. Done using dovetail locks, possibly T-shaped/rectangular bars, also using a footplate and secure probe holders so data is repeatable and reliable.
5. 1/22/2026	Came back to further investigate GPR	1 hr		investigated block diagrams and electric system diagrams for GPR. Devised a way to replace necessary parts for affordability and FMCW application.
6. 1/23/2026	Met with and searched for part links	25 min		went over the parts list sent with to understand what team had vs what team needed to buy vs what needed to buy themsleves. Per request, then began the process of finding links for all equipment team didn't have.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 1/23/2026	Worked on documentation for the 3 ideas that the team came up with	25 min		went over the solutions found, and formatted for the documentation. Transcribed discussion on final solution.
2. 1/25/2026	Started learning python and installed necessary components	2 hr		watched tutorials on how to program in python, how to download python software, and additional mathematical extensions needed. Prepared the folder, main, and virtual environment for the project.
3. 1/25/2026	Further researched python usage in scientific fields and began	45 min		watched videos on mathematical ways to use python and experimented.
4. 1/26/2026	Diarama research	1 hr 50 min		identified dimension requirements and found links for all necessary items to build a sandbox diorama.
5. 1/26/2026	Team zoom meeting	45 min		had an online team meeting, discussing potential design plans. Received constructive criticism and began physically mapping out what we physically want to make.
6. 1/27/2026	Research and sourcing	2 hr 45 min		researched GPR build designs to better understand how to best construct FMCW technology design and completed sourcing where to find necessary parts for purchase. Completed a parts list with links, required completing the design plan for our build.

PLAN OF WORK LOG

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Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 1/27/2026	Team zoom meeting	40 min		had an online team meeting, discussed the design plan and work expectations moving forward. Answered all questions and discussed all potential problems together.
2. 1/28/2026	General link and work documentation tidy-up	20 min		described all undocumented work on the Work Log and transferred links from my tabs to the Work Log.
3. 2/8/2026	Worked on modeling prototype	15 min		began modeling on Inventor.
4. 2/10/2026	Modeling prototype	15 min		continued modeling.
5. 2/17/2026	Held a team meeting	25 min		had an important meeting discussing deadlines for portfolio submission and priorities moving forwards in order to be able to finish everything in time for state.
6. 2/17/2026	Modeling	1 hr		and reviewed model on Inventor and began redesigning the "Sled" portion.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 2/20/2026	Modeling	4hr 30min		and completed designing of male-dovetail/handle part and ERT probe rail part in Inventor. Sent files to be 3D printed.
2. 2/21/2026	Transferred all of our current work log to the official TSA work log	2hr 5 min		moved the items in the current Google Sheets work log to the official TSA work log and tidied up the work log.
3. 2/21/2026	Documentation	3 hr		fixed documentation formatting and cited sources.
4. 2/22/2026	Modeling and Sketching	4 hr		continued modeling,
5. 2/22/2026	Documentation	1 hr		viewed the scoring rubric and worked on documentation.
6. 2/22/2026	Documentation and Modeling	2 hr		and worked through modeling ideas for combination mechanisms and continued documentation diagram work.

PLAN OF WORK LOG

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 2/23/2026	Documentation	1 hr		worked on documentation writing.
2. 2/23/2026	Documentation	40 min		continued working on documentation.
3. 2/23/2026	Documentation	3 hr		Finalized documentation writing portions.
4. 2/24/2026	Documentation	1hr 15 min		organized and formatted document.
5. 3/1/2026	Documentation	1hr 55 min		worked on organizing and formatting the rest of the document.
6.				

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (studentinitials)	**Comments**
3/23/2026	researched on further developing an affordable GPR to improve upon the design from state	45 min		found sensors that would properally image depth and density as well as found a way to accurately track position in order to create a 3d heat map of things below
3/27/2026	looked into making own pcb as well as dimensions of possible sensors	30 mins		in efforts to control the size of the circuitry and have a cleaner look, we looked into designing out own pcb. Wasn't entiry a necessary or viable solution for our project
4/1/2026	decided to develope the wenner array further	45 min		did a decision matrix, on how to proceed with the project. Decided to invest time into one idea rather than multiple so we have the best possible outcome.
4/7/2026	researched developing wenner further	1 hour		found better connections to rod, rods, processor, breakout module
4/11/2026	found parts	1 hour		found parts that could be used to make the enhanced wenner array

Advisor signature

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (studentinitials)	**Comments**
4/14/2026	ordered parts	25 min		sourced the parts the make the wenner array from a reputable site, that would arrive in a timely manner
5/1/2026	recieved parts	30 min		tested and checked all parts
5/7/2026	designed the circuitry	6 hr		brought all the parts together into one
5/14/2026	reworked circuitry	3 hr		work out small kinks in the design
5/16/2026	designed housing for the system	2 hr		designed a rough housing for testing purposes
5/25/2026	final housing for the system	30 mins		added final touches and features
5/27/2026	Desgined the board	90 min		designed the poster for the Nationals Presentation

Advisor signature _____

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (studentinitials)	**Comments**
5/28/2026	Documentation	1 hr		added updated information onto the documentation as well as updated images
5/29/2026	Documentation	3 hrs		continuing adding information to the documentation and added updated graphics and charts
5/30/2028	Documentation	90 mins		Worked on formating and aesthetics as well
6/1/2026	programming	2 hrs		worked on programming and learning new approaches to programming
6/2/2026	troubleshooting	1 hr		worked on finding bugs in the programming, and fixing any and all issues

Advisor signature _____

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